

1. NumPy (Numerical Python)

- Array creation
- Array indexing & slicing
- Broadcasting
- Vectorized operations
- Mathematical/statistical functions
- Reshaping arrays
- Stacking arrays
- Boolean masking
- Random number generation
- Performance (vs lists)

2. Pandas

- Reading/Writing data (CSV, Excel)
- DataFrame & Series basics
- Filtering rows
- Handling missing values
- GroupBy operations
- Merging/Joining
- Sorting
- Creating calculated columns
- Pivot tables
- Date-time operations

3. Matplotlib

- Line plot
- Bar chart
- Pie chart
- Histogram
- Scatter plot
- Subplots
- Labels, titles, legends
- Styling plots
- Saving figures

4. Seaborn

- Distribution plots (hist, kde)
- Cat plots (bar, count, box, violin)
- Relational plots (line, scatter)
- Heatmaps
- Pairplots
- Using hue, style, size
- Themes & styling

Exercise 1: Load Excel File Using Pandas

File: sample_exercise_1.xlsx

Task: Load data and display first 5 rows.

Solution:

```
import pandas as pd
```

```
df = pd.read_excel("sample_exercise_1.xlsx")  
print(df.head())
```

Explanation

- read_excel() loads the file
 - head() shows first 5 rows
-

Exercise 2: Calculate Mean of a Column (NumPy)

File: sample_exercise_2.xlsx

Task: Calculate mean of Value1.

Solution:

```
import pandas as pd  
import numpy as np
```

```
df = pd.read_excel("sample_exercise_2.xlsx")  
mean_val = np.mean(df["Value1"])  
print(mean_val)
```

Exercise 3: Add New Column (Value1 + Value2)

File: sample_exercise_3.xlsx

Solution:

```
df["Total"] = df["Value1"] + df["Value2"]  
print(df)
```

Exercise 4: Filter Rows (Value1 > 10)

File: sample_exercise_4.xlsx

Solution:

```
filtered = df[df["Value1"] > 10]
```

```
print(filtered)
```

Exercise 5: Group By and Sum

File: sample_exercise_5.xlsx

Add a new column Group = ID % 2 and group by.

Solution:

```
df["Group"] = df["ID"] % 2
result = df.groupby("Group")[["Value1", "Value2"]].sum()
print(result)
```

Exercise 6: Line Plot Using Matplotlib

File: sample_exercise_6.xlsx

Solution:

```
import matplotlib.pyplot as plt
```

```
plt.plot(df["ID"], df["Value1"])
```

```
plt.xlabel("ID")
```

```
plt.ylabel("Value1")
```

```
plt.title("ID vs Value1")
```

```
plt.show()
```

Exercise 7: Bar Plot Using Seaborn

File: sample_exercise_7.xlsx

Solution:

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
sns.barplot(x="ID", y="Value2", data=df)
```

```
plt.show()
```

Exercise 8: Create NumPy Array from DataFrame

File: sample_exercise_8.xlsx

Solution:

```
arr = df[["Value1", "Value2"]].to_numpy()
print(arr)
```

Exercise 9: Heatmap of Correlation Matrix

File: sample_exercise_9.xlsx

Solution:

```
corr = df.corr()
sns.heatmap(corr, annot=True)
plt.show()
```

Exercise 10: Scatter Plot (Value1 vs Value2)

File: sample_exercise_10.xlsx

Solution:

```
plt.scatter(df["Value1"], df["Value2"])
plt.xlabel("Value1")
plt.ylabel("Value2")
plt.title("Scatter Plot")
plt.show()
```